

## CRESTON, BC, Canada

WMO: 717700

Lat: 49.0817N

Lon: 116.5007W

Elev: 2120

StdP: 13.60

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 7.1        | 12.0 | -5.0                          | 4.5 | 9.5  | 1.9 | 6.6 | 14.3 | 16.6                  | 26.9 | 15.1 | 26.4 | 6.1                   | 70   | 0.515 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 24.5                   | 90.7            | 66.7 | 87.2 | 65.3 | 83.8 | 63.9 | 69.3                | 85.8 | 67.1 | 83.0 | 65.1 | 80.5 | 5.0                  | 300  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.6                            | 95.2 | 77.7 | 61.3 | 87.5 | 74.0 | 59.2 | 81.2 | 71.9 | 34.7          | 86.3 | 32.8 | 83.1 | 31.2 | 81.1 | 79.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |  |  |
| 15.6              | 13.3 | 11.4 | 2.6                        | 96.2 | 6.2                | 2.7 | -1.8                                               | 98.2 | -5.4       | 99.8 | -8.9       | 101.4 | -13.4      | 103.3 |  |  |
|                   |      |      | 1.5                        | 73.8 | 6.1                | 2.5 | -3.0                                               | 75.6 | -6.5       | 77.1 | -10.0      | 78.5  | -14.5      | 80.3  |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 48.1   | 29.2  | 31.8  | 39.2  | 47.2  | 56.2  | 61.9  | 70.5  | 69.4  | 59.4  | 46.4  | 35.8  | 28.5 |
|                                                                                  | DBStd       | 16.16  | 7.95  | 7.23  | 6.39  | 5.93  | 6.97  | 6.63  | 6.31  | 5.93  | 6.65  | 6.41  | 6.81  | 7.60 |
|                                                                                  | HDD50       | 2880   | 643   | 511   | 335   | 123   | 20    | 2     | 0     | 0     | 6     | 147   | 426   | 667  |
|                                                                                  | HDD65       | 6622   | 1108  | 931   | 799   | 534   | 285   | 137   | 24    | 27    | 192   | 577   | 876   | 1132 |
|                                                                                  | CDD50       | 2170   | 0     | 0     | 2     | 39    | 211   | 358   | 634   | 603   | 288   | 35    | 0     | 0    |
|                                                                                  | CDD65       | 436    | 0     | 0     | 0     | 0     | 11    | 43    | 194   | 165   | 23    | 0     | 0     | 0    |
|                                                                                  | CDH74       | 4355   | 0     | 0     | 0     | 6     | 173   | 449   | 1940  | 1549  | 237   | 1     | 0     | 0    |
|                                                                                  | CDH80       | 1622   | 0     | 0     | 0     | 0     | 33    | 129   | 809   | 595   | 56    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.3    | 3.5   | 3.9   | 4.7   | 5.2   | 4.8   | 4.6   | 4.6   | 4.4   | 4.1   | 3.8   | 3.8   | 3.7  |
| Precipitation                                                                    | PrecAvg     | 22.8   | 2.5   | 1.5   | 1.7   | 1.5   | 2.2   | 2.3   | 1.2   | 1.1   | 1.2   | 1.7   | 2.6   | 2.8  |
|                                                                                  | PrecMax     | 32.8   | 4.9   | 3.3   | 3.6   | 3.1   | 6.5   | 5.4   | 2.3   | 2.6   | 3.8   | 6.3   | 6.7   | 5.5  |
|                                                                                  | PrecMin     | 18.1   | 1.0   | 0.3   | 0.0   | 0.6   | 0.9   | 0.6   | 0.0   | 0.1   | 0.0   | 0.0   | 0.9   | 1.2  |
|                                                                                  | PrecStd     | 3.9    | 0.9   | 0.7   | 1.0   | 0.6   | 1.2   | 1.1   | 0.7   | 0.6   | 0.9   | 1.2   | 1.3   | 1.1  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 45.9  | 49.5  | 61.7  | 73.9  | 84.1  | 89.6  | 95.4  | 94.1  | 86.0  | 69.6  | 54.6  | 45.7 |
|                                                                                  |             | MCWB   | 41.7  | 42.7  | 48.5  | 56.6  | 63.0  | 65.9  | 68.5  | 67.7  | 62.6  | 57.5  | 49.9  | 41.2 |
|                                                                                  | 2%          | DB     | 42.2  | 44.8  | 55.8  | 67.1  | 79.0  | 83.5  | 91.9  | 90.1  | 80.3  | 63.6  | 49.7  | 42.3 |
|                                                                                  |             | MCWB   | 38.8  | 40.1  | 45.8  | 52.5  | 59.7  | 64.3  | 67.4  | 65.8  | 61.4  | 52.9  | 45.6  | 39.4 |
|                                                                                  | 5%          | DB     | 39.7  | 42.3  | 51.6  | 62.5  | 74.2  | 79.1  | 88.3  | 86.6  | 76.0  | 59.4  | 46.8  | 39.6 |
|                                                                                  |             | MCWB   | 37.3  | 38.3  | 43.6  | 49.8  | 57.3  | 62.3  | 66.3  | 64.4  | 59.6  | 50.8  | 43.9  | 37.4 |
|                                                                                  | 10%         | DB     | 37.6  | 39.9  | 48.3  | 58.3  | 69.4  | 74.9  | 84.6  | 83.0  | 71.5  | 55.8  | 44.6  | 37.1 |
|                                                                                  |             | MCWB   | 35.8  | 36.8  | 41.7  | 47.6  | 55.0  | 59.8  | 65.1  | 63.2  | 57.8  | 49.0  | 42.3  | 35.5 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 42.6  | 44.5  | 50.1  | 57.7  | 64.8  | 69.4  | 73.5  | 71.3  | 65.1  | 58.6  | 50.9  | 42.8 |
|                                                                                  |             | MCDB   | 45.2  | 47.0  | 57.5  | 71.6  | 80.9  | 85.5  | 90.3  | 88.3  | 80.7  | 66.2  | 53.5  | 44.4 |
|                                                                                  | 2%          | WB     | 39.5  | 41.0  | 46.8  | 53.6  | 61.2  | 66.0  | 70.5  | 68.1  | 62.6  | 54.8  | 46.9  | 40.2 |
|                                                                                  |             | MCDB   | 41.5  | 43.8  | 54.2  | 65.1  | 75.3  | 80.1  | 86.6  | 85.4  | 76.7  | 61.6  | 48.8  | 42.0 |
|                                                                                  | 5%          | WB     | 37.7  | 38.9  | 44.6  | 50.7  | 58.7  | 63.3  | 68.2  | 66.0  | 60.7  | 52.1  | 44.7  | 37.9 |
|                                                                                  |             | MCDB   | 39.4  | 41.4  | 50.2  | 60.4  | 71.4  | 76.7  | 84.0  | 82.9  | 73.7  | 57.4  | 46.4  | 39.4 |
|                                                                                  | 10%         | WB     | 36.0  | 37.1  | 42.5  | 48.4  | 56.4  | 61.0  | 66.0  | 64.3  | 58.7  | 49.8  | 42.4  | 35.8 |
|                                                                                  |             | MCDB   | 37.3  | 39.3  | 47.2  | 56.7  | 67.6  | 72.2  | 81.2  | 79.7  | 70.1  | 55.0  | 44.2  | 37.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.1   | 11.0  | 14.7  | 18.4  | 20.9  | 20.3  | 24.5  | 24.5  | 21.1  | 15.5  | 9.1   | 7.4  |
|                                                                                  |             | MCDBR  | 9.9   | 12.6  | 20.1  | 25.7  | 28.0  | 26.9  | 29.1  | 29.2  | 27.4  | 20.1  | 11.6  | 9.4  |
|                                                                                  |             | MCWBR  | 7.5   | 8.6   | 12.3  | 14.4  | 14.8  | 14.2  | 14.0  | 13.6  | 13.6  | 11.8  | 8.4   | 7.2  |
|                                                                                  | 5% WB       | MCDBR  | 9.6   | 11.7  | 18.5  | 24.0  | 25.7  | 25.1  | 27.1  | 27.1  | 25.4  | 18.2  | 10.6  | 8.8  |
| MCWBR                                                                            |             | 7.4    | 8.4   | 11.9  | 14.1  | 14.4  | 13.9  | 14.4  | 13.5  | 13.1  | 11.1  | 8.3   | 7.1   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.267  | 0.276 | 0.298 | 0.368 | 0.374 | 0.371 | 0.366 | 0.379 | 0.343 | 0.313 | 0.287 | 0.262 |      |
|                                                                                  | taud        | 2.323  | 2.304 | 2.293 | 2.204 | 2.340 | 2.394 | 2.417 | 2.386 | 2.473 | 2.539 | 2.480 | 2.356 |      |
|                                                                                  | Ebn at Noon | 253    | 279   | 291   | 278   | 281   | 282   | 282   | 272   | 273   | 265   | 247   | 239   |      |
|                                                                                  | Edh at Noon | 22     | 28    | 34    | 42    | 38    | 36    | 35    | 35    | 29    | 23    | 19    | 18    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 297    | 591   | 881   | 1358  | 1696  | 1784  | 2156  | 1803  | 1265  | 725   | 328   | 235   |      |
|                                                                                  | RadStd      | 43     | 76    | 124   | 129   | 141   | 169   | 166   | 142   | 136   | 98    | 47    | 32    |      |

## Historical Trends

|                              |       | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (47) Regional (18 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page